

TrES-3b

R-0407

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250812 · created 2026-07-19T05:17:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460899.7096 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.057
Transit depth (Rp/Rs)^2	1.83 +/- 1.54 [%]
Orbital Inclination (inc)	79.84 +/- 2.9
Airmass coefficient 1 (a1)	0.44 +/- 0.19
Airmass coefficient 2 (a2)	0.8 +/- 0.8
Scatter in the residuals of the lightcurve fit is	45.69 %
Best Comparison Star	#3 - [623, 192]
Optimal Aperture	1.58
Optimal Annulus	13.17
Transit Duration (day)	0.026 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.057 vs 0.16309 (deviation 0.4 σ)
Duration fit vs published	0.026 d vs 0.0567 d (deviation 0.92 σ)
Mid-time O–C	0.5 min
Depth SNR	1.9
Residual scatter	45.69 %

Light curve

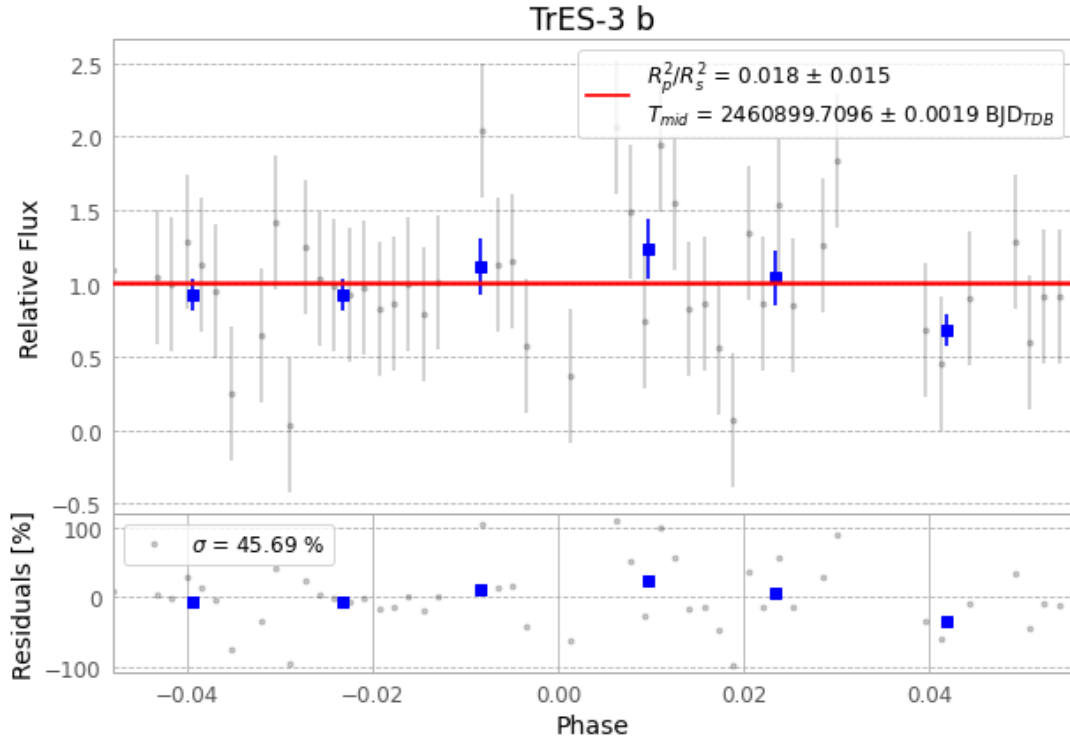


Figure 1 — FinalLightCurve_TrES-3 b_2025-08-11.png (SHA-256 6bf1c1d5707e3898...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df65e1dd0811b785...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

66 FITS frames (43 MB), TRES-3250812032717.FITS → TRES-3250812064215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250812.log (cc66d8bfbd8...), FinalLightCurve_TrES-3 b_2025-08-11.png (6bf1c1d5707e...), FinalLightCurve_TrES-3 b_2025-08-11.csv (a62e750b2f54...)

Compute resources

Wall time 225.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 05:17Z.